

HOANG MINH NGUYEN

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EDUCATION

Florida State University

Ph.D., Financial Mathematics. GPA: 3.92

Tallahassee, FL

December 2024

The University of Mississippi

B.S., Mathematics. B.B.A., Managerial Finance. GPA: 3.91 - Summa Cum Laude, Honors Scholar

Oxford, MS

May 2019

WORK EXPERIENCE

Meta Platforms

Menlo Park, CA

Research Scientist

Jan 2025 - Present

- Designed and constructed large language model (LLM)-based system to analyze and detect comment sentiments (“vibes”) on Facebook
- Built and deployed end-to-end pipelines with multi-task, text-based machine learning models to capture fine-grained sentiment nuances, addressing LLM capacity constraints and the enormous scale of Facebook traffic
- Conducted user testing and integrated models into Facebook’s comment ranking system, increasing authenticity by 4.5% and reducing bad vibe prevalence by 22.5%, resulting in significant gains in engagements metrics

Meta Platforms

Menlo Park, CA

Machine Learning Software Engineer PhD Intern

May 2024 - August 2024

- Assisted the Modern Recommendation System AI Team in building multimodal content understanding models, with a focus on audio understanding and representations
- Implemented the state-of-the-art model for audio understanding with tokenization, quantization methods used in research and vision transformer - based audio encoder to produce audio embeddings; built a full pipeline for training and evaluating the model in audio acoustic events prediction task, and in multimodal fusion with visual and text inputs
- Constructed a well-balanced internal dataset for audio pretraining, leading to an improvement of ~20% in internal audio understanding evaluation tasks on public data AudioSet, containing audio events of 10-second Youtube clips, and the best internal performance in multimodal Instagram Reels topic prediction task with text, visual and audio fusion

Wells Fargo

Remote

Quantitative Analytics Intern, Artificial Intelligence Modeling Team

June 2021 - August 2021

- Conducted research on Generative Adversarial Networks (GANs) to solve the problem of imbalanced datasets; implemented different GAN types using Tensorflow to generate realistic synthetic data for given datasets
- Experimented, adjusted GANs by manipulating the neural network structures and tested the performances of GANs on machine learning classifiers, including Logistic Regression, Random Forest, XGBoost and Neural Networks; successfully increased the recall rate of the classifiers by 2-3% on multiple imbalanced data problems

Reinsurance Group of America

Remote

Actuarial Intern, Global Research & Data Analytics Department

June 2020 - August 2020

- Built VBA and R programs to query 40 million entries of U.S. death data; sped up the search tool of the company by approximately 10 times while working with a large database
- Utilized SQL tools to analyze and detect the top diseases and health disorders with potential high future claim costs using the company’s historical claims and patients’ diagnosis data

Florida State University

Tallahassee, FL

Graduate Teaching Assistant, Department of Mathematics

August 2019 - present

- Perform assistant teaching duties, including lecturing and grading for Calculus I, Discrete Mathematics, College Algebra and Precalculus
- Mentor a class of 100+ students and deliver a range of teaching and assessment activities including tutorials at undergraduate mathematical level

SKILLS

- Programming languages: Python (ML libraries: Pytorch, Tensorflow, Scikit-learn), R, C/C++, SQL, Matlab
- Research expertise: Generative AI, Multimodal AI, Diffusion Models, Computer Vision, Foundational Models
- Strong quantitative and analytical skills in interpreting both the numerical and theoretical results of a model

RESEARCH EXPERIENCE

Publications

Hoang M. Nguyen, Xuefeng Gao, and Lingjiong Zhu. "Wasserstein Convergence Guarantees for a General Class of Score-Based Generative Models." Journal of Machine Learning Research, 2025. [[paper](#)] | [[Github](#)]

- Proposed and implemented new stochastic differential equations (SDEs) with Pytorch on the current framework of score-based diffusion models, the state-of-the-art diffusion models for image generations
- Analyzed the convergence rates of Noise Conditional Score Networks and Denoising Diffusion Probabilistic Models with the existing and newly proposed SDEs; demonstrated the alignment between the theoretical guarantees and the empirical metrics of the generated fake images on CIFAR-10 dataset

Hoang M. Nguyen, Mert Gurbuzbalaban, Xicheng Zhang, Lingjiong Zhu. "Anchored Langevin Algorithms." 2025 [[arXiv](#)] | [[Github](#)]

- Developed new Gradient Descent Langevin Dynamics by modifying the loss function using smoothing methods and scaling the gradients, specifically targeting non-differentiable or non-convex functions in machine learning applications such as ReLU activation, Lasso, SCAD and MCP regularizers
- Fully designed experiments to verify the behaviors of the above algorithms using distributions sampling, Logistic Regression and Neural Networks with Anchored Langevin Gradient Descent algorithms as the optimizer
- Obtained convergence with up to 90% average accuracy for practical classification problem and non-convex sampling problems where the existing Gradient Descent Langevin Dynamics fails to converge

Hoang M. Nguyen, Satya N. Shukla, Qiang Zhang, Hanchao Yu, Sreya D. Roy, Taipeng Tian, Lingjiong Zhu, Yuchen Liu. "BRIDLE: Generalized Self-supervised Learning with Quantization." 2025 [[arXiv](#)]

- Developed self-supervised encoder pretraining framework for audio, image, and video classification tasks, incorporating residual quantization into bidirectional training process, enhancing representation quality
- Achieved state-of-the-art results on audio classification benchmarks, and demonstrated competitive performance on image and video datasets

Doctoral Dissertation, Florida State University [[dissertation](#)] 2025

Langevin Dynamics and Diffusion Models: Sampling in Machine Learning and Generative AI

Honors Thesis, The University of Mississippi [[thesis](#)] 2019

Portfolio Optimization Methods: The Mean-Variance Approach and the Bayesian Approach

CERTIFICATIONS & PROFESSIONAL EXAMS

- **Coursera Specializations:** Machine Learning, Deep Learning, Natural Language Processing (DeepLearning.AI)
- **Probability Test**, certified by Society of Actuaries September 2018
- **Financial Mathematics**, certified by Society of Actuaries August 2018
- **Chartered Financial Analyst (CFA) level 1**, certified by CFA Institute June 2018

HONORS, INVOLVEMENT & LEADERSHIP EXPERIENCE

- Alfred Hume Award for most outstanding senior at the Department of Mathematics 2019
- Bickerstaff Scholarship for most outstanding junior at the Department of Mathematics 2018
- President of the International & American Student Alliance IASA, August 2018 - May 2019
 - Expanded the three-year old organization into the largest international community on campus with a board of 17 directors and approximately 1000 members at University of Mississippi
 - Obtained at least \$2000 every semester to facilitate substantial events and various programs to provide the international students with academic support and experience
- Rank 298th scorer of William Lowell Putnam Mathematical Competition 2016

CUSTOMER SERVICE EXPERIENCE

Community Assistant, The University of Mississippi - Department of Housing August 2016 - August 2018

- Facilitated social, academic, and personal development for a community of around 30 residents in the residence hall
- Organized successfully and supported community events, designed and implemented experiences to meet established goals of Department of Housing